

Media Bias ISM

Regressions Compendium — All Specifications, Equations, and Results

Overview

This document consolidates every regression specification estimated in the ISM Media Bias project on the cross-firm daily/event panels (2015–2025 S&P 500 firms). The VAR / Granger time-series track for individual tickers is documented separately and is intentionally not included here. For each regression we record: (i) motivation, (ii) the equation, (iii) the estimation sample, (iv) the coefficient table with HC1 robust standard errors, and (v) an interpretation paragraph. A glossary, a cross-regression summary, an identification and robustness section, the open to-dos, and a variables index follow at the end.

Glossary

Project-specific terms.

Stance / bias

The signed quantity $p^+ - p^-$ assigned to an article–firm pair by NewsMTSC, where p^+ and p^- are model probabilities for positive and negative stance toward the target firm. Bounded in $[-1, +1]$. Averaged across articles to form a daily firm-day score bias_{it} .

NewsMTSC

Hamborg and Donnay (2021) target-dependent stance classifier. Produces (p^+, p^-, p^0) per article–firm pair. “Target-dependent” means the score is specific to a named firm, not a generic article sentiment.

Confidence threshold (τ)

Minimum $\max\{p^+, p^-, p^0\}$ required to retain an article–firm output. Default $\tau = 0$ (keep all). Sensitivity reported at $\tau \in \{0.0, 0.6, 0.9\}$.

Relevance classifier

DeBERTa-v3-base binary model that filters titles to those materially about the target firm. Used as a pre-stage before NewsMTSC stance scoring.

Post indicator

$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01]$. Marks the 2020 recession breakpoint. Pre-period: 2015–2019. Post-period: 2020–2025.

Firm fixed effect (α_i)

One intercept per firm, absorbs every time-invariant firm characteristic (sector, size, baseline coverage).

Time fixed effect (δ_t)

One intercept per day (or per month for some specs), absorbs every firm-invariant time shock (Fed days, COVID waves, election cycles).

Two-way FE

Both α_i and δ_t in the same model. Stricter than one-way FE: the Post coefficient β is identified only from within-firm–within-day deviations.

Econometric and statistical terms.

OLS

Ordinary Least Squares. Estimator that minimizes squared residuals.

DiD (Difference-in-Differences)

Compares the pre/post change in a treated group with the pre/post change in a control group. With universal treatment (everyone hit by the shock at the same time), DiD reduces to a within-firm pre/post comparison.

Parallel trends

For DiD to identify a causal effect, treated and control groups must have parallel pre-shock outcomes. Tested by examining pre-period event-study coefficients.

Event study

Generalizes DiD by replacing a single Post dummy with one dummy per period relative to the event. Coefficients β_k at event-time k trace out the dynamic path.

Reference bin

In an event study, the period whose dummy is omitted to avoid collinearity. All β_k are interpreted relative to this reference. Standard choice $k = -1$.

HC1 robust standard error

Heteroscedasticity-consistent SE with small-sample correction $n/(n-K)$. Default uncertainty estimator when residuals are not homoscedastic.

Day-clustered SE

Cluster-robust SE with clusters defined at the day level. Robust to arbitrary within-day correlation across firms (useful when all firms share daily macro shocks).

Within R^2

R^2 after the fixed-effects within-transform. Measures variance explained within entities (firms), not overall variance.

t -statistic, p -value, 95% CI

$t = \hat{\beta}/\widehat{SE}$; $p = 2\Phi(-|t|)$; CI = $[\hat{\beta} \pm 1.96\widehat{SE}]$. Coefficient is significant at 5% if CI excludes 0.

F -test

Tests joint significance of a group of coefficients. Used here to check whether 4,017 daily fixed effects together explain variation beyond what one-way firm FE explains.

BLUE / Gauss-Markov

An estimator is BLUE if errors have zero mean, constant variance, and no autocorrelation. Robust SEs are used when these fail.

Breusch-Pagan, Jarque-Bera, VIF, Durbin-Watson

Standard residual diagnostics (heteroscedasticity, normality, collinearity, serial correlation).

Project Motivation

The 2020 recession is a clean, universal, exogenous shock to every large-cap US firm. It removes the usual hard problem of finding a valid control group: every firm is treated simultaneously. That makes it a natural setting for testing pre/post structural changes in firm-level outcomes.

The project asks a chain of three questions:

1. **RQ1.** Did the *tone* of media coverage of S&P 500 firms shift after Jan 2020?
2. **RQ2.** Did firm-level *returns* shift after Jan 2020 beyond what market co-movement explains?
3. **RQ3.** If both shifted, did the *relationship* between media tone and returns strengthen post-shock?

Data and Variable Construction

Corpus and sampling. The raw scrape contains 4,706,589 firm–article links across 30 S&P 500 firms, 2015–2025. We filter to articles whose titles explicitly contain the company name (not ticker, to avoid Apple-the-fruit / Visa-the-immigration problems), de-duplicate by normalized title within firm, and drop firms with more than three years of $\leq 3,000$ articles. The retained set is 26 firms and 690,586 firm–article links.

Daily stance series. For each retained article a of firm i on date t , with NewsMTSC probabilities (p_a^+, p_a^-, p_a^0) :

$$s_a = p_a^+ - p_a^-, \quad \text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a.$$

$|\mathcal{A}_{it}|$ is the firm-day article count, also retained as a control vol_{it} . At $\tau = 0$ the RQ1 panel has 63,200 firm-day observations.

Daily returns and market controls.

$$\text{return}_{it} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}, \quad \text{sp500_return}_t = \frac{P_t^{\text{SP500}} - P_{t-1}^{\text{SP500}}}{P_{t-1}^{\text{SP500}}}, \quad \text{vix}_t = \text{CBOE VIX close on day } t.$$

The RQ2 panel has 62,667 firm-trading-day observations.

Shock indicator. $\text{Post}_t = \mathbb{1}[t \geq 2020-01-01]$, pre: 2015-2019, post: 2020-2025.

Table 1: *
Panel sizes across the three research questions

Track	Observations	Firms	Date / frequency
RQ1 (daily stance)	63,200 firm-days	26	2015-01-01 to 2025-12-31 daily
RQ2 (daily returns)	62,667 firm-days	26	2015-12-01 to 2025-11-24 daily
RQ3.1 (price-shock events)	69,248 / 64,112 events	26	2015-12-01 to 2025-11-24 daily
RQ3.2 (bias $\rightarrow$ return)	140 (overall)	26	2015-12-01 to 2025-11-24 daily

RQ1.1 — Two-Way FE (Firm + Day) on Stance

Motivation. Simplest pre/post test on stance. Add day fixed effects on top of firm fixed effects so that the Post coefficient is identified only from within-firm-within-day variation, net of every common shock that hit all firms on a given day.

Specification.

$$\text{bias}_{it} = \alpha_i + \delta_t + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it}.$$

Sample. $n = 63,200$ firm-days, 26 firms, 4,018 distinct trading days, $\tau = 0$.

Table 2: *
Table RQ1.1: Two-Way FE on stance (HC1 robust SE)

Variable	Coef.	SE (HC1)	t -stat	p -value	CI Low	CI High
Post	0.014130	0.013623	1.0372	0.2996	-0.012571	0.040831
Article Volume	-0.000970	0.000076	-12.7724	$< 10^{-37}$	-0.001119	-0.000822
Within R^2	0.001749					

Interpretation. The post-2020 within-firm shift in stance is positive (+0.014) but *not* significant ($p = 0.30$) after absorbing daily common shocks. Article volume is strongly negative and highly significant: one extra firm-day article is associated with a 0.001 drop in stance, consistent with negative-bias-driven coverage spikes. Within R^2 is small (0.0017): most variance in bias_{it} is firm-by-day idiosyncratic noise that two regressors cannot capture.

Pending: log-normalised volume

The volume control is currently included in *raw count* form. A log transform $\widetilde{\text{vol}}_{it} = \ln(1 + \text{vol}_{it})$ is the standard fix for right-skewed counts. It compresses the influence of high-coverage outliers (max 539 articles per firm-day) and converts the coefficient interpretation to “percent change in stance per percent change in coverage.” *Open to-do — not yet re-estimated.*

RQ1.2 — Same Spec, Day-Clustered SE

Motivation. HC1 assumes independent observations conditional on covariates. In a daily panel, two firms on the same day share macro shocks, so residuals can be correlated within day. Day-clustered SEs are robust to this dependence and are the preferred uncertainty estimate for daily panels.

Specification. Identical to RQ1.1, only the variance estimator changes:

$$\widehat{\text{Var}}(\hat{\theta}) = (\tilde{X}^\top \tilde{X})^{-1} \left(\sum_{g \in \text{days}} \tilde{X}_g^\top \hat{u}_g \hat{u}_g^\top \tilde{X}_g \right) (\tilde{X}^\top \tilde{X})^{-1}.$$

Table 3: *

Table RQ1.2: Day-clustered SE (4,018 clusters)

Variable	Coef.	SE (clustered)	t -stat	p -value	CI Low	CI High
Post	0.014130	0.013432	1.0520	0.2928	-0.012196	0.040456
Article Volume	-0.000970	0.000076	-12.8438	$< 10^{-37}$	-0.001119	-0.000822

Interpretation. Day-clustered SEs are virtually identical to HC1 SEs ($< 1\%$ deviation). Inference does not depend on the cluster assumption; the insignificance of Post is not an artifact of ignoring within-day correlation.

RQ1.3 — Joint F -Test for Day Fixed Effects

Motivation. The single Post dummy assumes the post-2020 shift is constant across all days. The day FE jointly span every individual daily shock. If they are not jointly significant, including them is overkill; if they are, the time variation matters and the static Post compresses dynamic signal.

Specification. F -test of $H_0 : \delta_1 = \delta_2 = \dots = \delta_T = 0$ in the two-way FE model.

Table 4: *

	Statistic	Value
	F -stat	1.1348
Table RQ1.3: Joint significance of day FE	df1 (time periods - 1)	4,017
	df2 (residual df)	59,181
	p -value	1.12×10^{-8}
	F critical (5%)	1.0383

Interpretation. Day FE are jointly significant at $p < 10^{-7}$. Daily common shocks matter and cannot be collapsed into a single Post indicator. This is the technical justification for the event-study specification in RQ1.4.

RQ1.4 — Monthly Event Study on Stance

Motivation. The static Post compresses everything post-Jan-2020 into one number. An event study replaces it with one coefficient per calendar month relative to Jan 2020, giving (i) a pre-trend test (months before should hover around zero) and (ii) the dynamic post-shock path.

Specification. Firm fixed effects + monthly event-time dummies, $k \in [-24, +24]$, reference $k = -1$ (Dec 2019):

$$\text{bias}_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \mathbb{I}[t \in \text{month}_k] + \gamma \text{vol}_{it} + \varepsilon_{it}.$$

$n = 23,439$ firm-day observations within the ± 24 -month window around Jan 2020.

Table 5: *

Table RQ1.4: Selected event-study coefficients (HC1 SE)

k (month)	Coef.	SE	t -stat	p -value	Comment
−24	0.04826	0.01791	2.694	0.007	pre-trend outlier (Jan 2018)
−12	0.02169	0.01805	1.202	0.229	pre-trend OK
−6	0.03176	0.01720	1.846	0.065	pre-trend mild outlier
−3	0.03442	0.01704	2.020	0.043	pre-trend mild outlier
−1	0	—	—	—	reference (Dec 2019)
0 (Jan 2020)	0.03103	0.01835	1.691	0.091	event month
+3	0.04215	0.01917	2.199	0.028	early post-shock
+6	0.03220	0.01825	1.764	0.078	mid post-shock
+12	0.06405	0.01880	3.408	$< 10^{-3}$	1-year post: significant
+18	0.05550	0.01878	2.955	0.003	sustained
+24	0.05039	0.01871	2.694	0.007	2-year post: sustained

Interpretation. Pre-period coefficients are mostly insignificant with a few mild outliers (Jan 2018, mid-2019) that do not form a systematic trend: parallel trends is reasonable. The event month $k = 0$ shows a marginal effect ($p = 0.09$). Effects emerge by $k = +3$ months (early COVID waves) and become *persistently* positive by $k = +12$, staying at +0.05 to +0.10 through $k = +24$. The static Post (RQ1.1) is insignificant because it averages an absent month-0 effect with a strong post-year-1 effect; once dynamics are allowed, the post-2020 stance shift is clear and persistent.

RQ1.5 — MLR / Gauss-Markov Diagnostics

Motivation. Verify the assumptions under which OLS is BLUE; flag any failures that require switching estimators or SEs.

Table 6: *

Table RQ1.5: Diagnostic tests on the two-way FE residuals ($n = 63,200$)

Assumption / Test	Result	Implication
Residual mean	-9.0×10^{-19}	✓ Orthogonality satisfied
Residual SD	0.278	✓ Non-trivial variation
Condition number	157.46	✓ Numerically stable
<i>Heteroscedasticity (Breusch-Pagan)</i>		
LM statistic	190.25	× Heteroscedasticity present
p -value	$< 10^{-42}$	HC1 robust SE used
<i>Normality (Jarque-Bera)</i>		
JB statistic	4,541.18	× Non-normality
Skewness, Kurtosis	−0.198 / 4.252	fat tails (typical for financial data)
<i>Multicollinearity (VIF)</i>		
Post, Article Volume	1.0004 each	✓ no collinearity
Max pairwise correlation	0.0189	✓ negligible
<i>Serial correlation (Durbin-Watson)</i>		
DW statistic	1.673	moderate; partly absorbed by FE

Interpretation. Two assumptions fail: residuals are heteroscedastic and non-normal. Both are handled by HC1 robust SEs and the large sample size ($n = 63,200$, well within CLT range). No collinearity. Residual mean is machine zero, confirming FE absorption. Day-clustered SEs (RQ1.2) virtually equal HC1 SEs, so within-day correlation is not large. Inference is sound.

RQ2.1 — Baseline Firm FE on Returns

Motivation. Mirror RQ1 on the returns side. If markets reacted to the same shock, we should see a within-firm pre/post shift in returns. Market and volatility controls strip out everything common to all firms.

Specification.

$$\text{return}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{sp500_return}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}.$$

$n = 62,667$ firm-trading-days. Pre-period firm means tightly clustered (range 0.00193, SD 0.00042), so the DiD parallel-trends assumption holds at the firm-mean level; no SCDiD fallback required.

Table 7: *
Table RQ2.1: Entity FE regression on returns (HC1 robust SE)

Variable	Coef.	SE	t -stat	p -value	CI Low	CI High
Post	−0.000344	0.000172	−2.0038	0.0451	−0.000681	−0.000008
S&P 500 Return	1.127576	0.010982	102.677	≈ 0	1.106052	1.149100
VIX	−0.000001	0.000020	−0.043	0.965	−0.000039	0.000038

Interpretation. Post-2020 daily returns are about 3.4 basis points lower on average than pre-2020, marginally significant ($p = 0.045$) after stripping out market index movement and volatility. The S&P 500 coefficient of about 1.13 is a textbook market beta. VIX is essentially zero once the index return is in (the two are highly correlated and the index absorbs the volatility-regime signal).

RQ3.1 — Event Study Around Extreme Price Shocks

Motivation. Test the reverse direction: do extreme *price* moves coincide with shifts in media coverage? An event study centred on $\pm 2\sigma$ price-shock days asks whether the 10 days of stance *after* a price shock differ from the 10 days *before*. Significant change is evidence of a price-to-media channel.

Specification. Paired-event t -test on the difference in mean stance between the 10-day windows before and after each $\pm 2\%$ daily return event, pooled across all firm-days that qualify.

Table 8: *
Table RQ3.1: Media change around extreme price shocks

Event type	Sample size	Mean media change (10d)	t -stat	p -value	Verdict
Positive price shock ($> +2\%$)	69,248	0.003291	2.840	0.002	significant
Negative price shock ($< -2\%$)	64,112	0.005229	4.374	$< 10^{-5}$	highly significant

Interpretation. Both directions show statistically significant media response to extreme price moves. The post-shock media-change is *positive* in both directions and *larger* for negative price shocks: tone reliably shifts up after a -2% day, plausibly reflecting analyst rebound coverage.

RQ3.2 — Bias-to>Returns at Multiple Horizons

Motivation. The canonical question of the project: does media bias predict subsequent returns, and does the relationship depend on the sign of the bias signal? Estimated separately at four horizons (1, 5, 14, 30 days) and three regimes (Overall, Positive bias subset, Negative bias subset).

Specification. For each (regime, horizon h) pair:

$$\text{return}_{i,t+h} = \alpha + \beta \widetilde{\text{bias}}_{it} + \varepsilon_{it},$$

where $\widetilde{\text{bias}}_{it}$ is the firm-demeaned bias and $h \in \{1, 5, 14, 30\}$ trading days ahead.

Table 9: *

Table RQ3.2: Bias-to-future-return at multiple horizons (selected rows)

Regime	Horizon	Coef. on bias	p -value	n	Verdict
Overall	1-day	−0.000682	0.471	140	n.s.
Overall	5-day	−0.000185	0.449	140	n.s.
Overall	14-day	−0.000979	0.114	140	marginal
Overall	30-day	−0.001352	2.55×10^{-19}	140	highly significant
Positive bias	1-day	−0.003587	0.584	75	n.s.
Positive bias	30-day	−0.002835	0.018	75	significant
Negative bias	1-day	+0.004406	0.364	61	n.s.
Negative bias	14-day	−0.002897	0.005	61	significant
Negative bias	30-day	+0.001670	0.660	61	n.s.

Interpretation. At short horizons (1–5 days) bias does not predict returns: the daily market is too noisy. At 14 days the negative-bias regime shows a significant negative effect (−0.29% further return over the next 14 days, $p = 0.005$). At 30 days the overall effect becomes *very* highly significant ($p < 10^{-18}$): a one-unit bias-demean increase predicts a −0.135% return 30 days out, consistent with a contrarian or mean-reversion channel. Caveat: n per regression is small (61–140) so the p -values rely on cross-event pooling, not firm-day granularity.

Cross-Regression Summary

Table 10: *

All eight regressions at a glance

ID	Spec	Outcome	Headline result	Verdict
RQ1.1	Two-way FE + Post + vol	stance	$\beta(\text{Post}) = +0.014, p = 0.30$	n.s. (static)
RQ1.2	+ day-clustered SE	stance	$\beta(\text{Post})$ same, $p = 0.29$	robust to clustering
RQ1.3	F -test on day FE	—	$F = 1.13, p = 1.1 \times 10^{-8}$	day FE matter
RQ1.4	Monthly event study	stance	Persistent post-shock uplift: $\beta_{+12} = +0.064, p < 10^{-3}$	sig. (dynamic)
RQ1.5	MLR diagnostics	residuals	BP $p < 10^{-42}$; JB $p \approx 0$; VIF ≈ 1	HC1 SE justified
RQ2.1	Firm FE + Post + market + VIX	return	$\beta(\text{Post}) = -0.000344, p = 0.045$	marginally sig.
RQ3.1	Event study around $\pm 2\sigma$ shocks	media change	$\mu = +0.0033, p = 0.002$ (up); $\mu = +0.0052, p < 10^{-5}$ (down)	sig. both directions
RQ3.2	Bias $\rightarrow$ return, 30-day overall	return	$\beta = -0.00135, p = 2.55 \times 10^{-19}$	highly sig.

Reading. Static pre/post tests (RQ1.1, RQ2.1) are weak or marginal because they compress five post-shock years into a single dummy. Event-study specs (RQ1.4, RQ3.1) tell the real story: stance shifts gradually and locks in higher by year 1; media reliably reacts to extreme price moves. RQ3.2 at 30-day horizon gives the headline causal-style number ($p < 10^{-18}$).

Identification and Robustness

Parallel trends. Pre-period coefficients in the RQ1.4 event study are mostly small and insignificant. For RQ2.1, pre-period firm means are tightly clustered (SD = 0.0004), so the DiD assumption holds at the firm-mean level; no SCDiD fallback

required.

Cluster vs HC1. Day-clustered SEs for RQ1.1 are within 1% of HC1 SEs, so within-day cross-firm correlation is not driving inference.

Heteroscedasticity and non-normality. Both present (RQ1.5). Handled with HC1 robust covariance and large-sample CLT inference; reported p -values use the two-sided normal approximation $p = 2\Phi(-|t|)$.

Day FE absorption. The F -test (RQ1.3) confirms day FE are jointly significant; they are the default in RQ1. They are not currently used in RQ2.1; the same two-way FE upgrade is on the to-do list.

Confidence threshold τ . Default $\tau = 0$. Sensitivity at $\tau \in \{0.6, 0.9\}$ is available via the RQ1 pipeline.

Relevance filtering. Before NewsMTSC stance scoring, articles pass through a DeBERTa-v3-base relevance classifier with operating point chosen to satisfy precision ≥ 0.90 (ensemble current best: $P = 0.944$, $R = 0.810$ on a 187-article held-out test set).

Open To-Dos

1. **Log-normalise volume.** Replace vol_{it} with $\ln(1 + \text{vol}_{it})$ in the RQ1 specifications (RQ1.1, RQ1.2, RQ1.4). Compresses high-coverage outliers and converts the volume coefficient to an elasticity. *Not yet re-estimated.*
2. **Two-way FE on returns (RQ2).** Add day or month FE to RQ2.1 to mirror RQ1 and confirm the static Post estimate survives absorbing common time shocks. *Pipeline supports it; not yet run.*

Variables Index

Table 11: *
All symbols used in this document

Symbol	Meaning	First appears
i	Firm index ($i \in \{1, \dots, 26\}$)	Data section
t	Date index (daily)	Data section
α_i	Firm fixed effect	RQ1.1
δ_t	Day (or month) fixed effect	RQ1.1
β	Main coefficient on Post indicator	RQ1.1
β_k	Event-study coefficient at relative time k	RQ1.4
$\gamma, \gamma_1, \gamma_2$	Control-variable coefficients	RQ1.1
ε_{it}	Idiosyncratic error term	RQ1.1
bias_{it}	Daily firm-day stance score (NewsMTSC)	Data section
s_a	Article-level stance: $p_a^+ - p_a^-$	Data section
p_a^+, p_a^-, p_a^0	NewsMTSC probabilities (pos / neg / neu)	Data section
τ	NewsMTSC confidence threshold (0, 0.6, 0.9)	Glossary
vol_{it}	Article count for firm i on day t	RQ1.1
Post_t	$\mathbb{I}[t \geq 2020-01-01]$	Data section
return_{it}	Daily simple return for firm i	Data section
sp500_return_t	Daily S&P 500 index return	Data section
vix_t	CBOE VIX close on day t	Data section
n	Estimation sample size	all regressions
K	Number of regressors	Glossary
$\widehat{SE}$	Standard error of an estimated coefficient	Glossary
$\widehat{\text{Var}}$	Estimated variance-covariance matrix	Glossary

Source files. RQ1 pipeline: `rq/rq-1-pipeline.py`. RQ2 pipeline: `rq/rq-2-pipeline.py`. MLR diagnostics: `rq/mlr_assumptions_check.py`. STMA / RQ3 anomaly detection: `stma/anomaly_det/`. *Result files.* `results/rq1/`, `results/rq2/`, `stma/anomaly_det/event_study/`, `stma/visualizations/impact_analysis/`. *Time-series VAR / Granger track is documented separately in VAR/ and is not included in this compendium.*